Atmospheric entry

antocciphysics | TEACHER KEY | CPA

Learning target: Use energy and geometry to discuss entry heating without claiming a guaranteed outcome.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 7; 3 course days later.*
$\Delta U = +20$ J.

02. Explain a worked example (worked)
Branches 700 and 300 J; could measure temperature change, emitted radiation or air heating with appropriate models.

03. Look and explain (model)
Smaller sphere has greater area per mass; stronger surface exposure per kg, all else equal.

04. Build the solution (practice)
Ratio 5; no, energy partitions among air, radiation, deformation and the object.

05. Use the idea (reason)
Entry angle/speed profile, density/strength, fragmentation, shape, heat-transfer rate, atmosphere; accept relevant factors.

06. Choose a method (mixed) *Review: Unit 4, Day 20; 12 course days later.*
No; it lacks 4 J of the required energy and remains bound under the model.

07. Exit check (exit)
Larger has half area per mass.

08. STRETCH: Apply it another way (stretch)
200 and 600 J; need mass, material, shape, heat transfer, ablation and changing conditions.

Teaching note: Science reference: NASA NTRS, “2.5 Meteorite Recovery” (20200000049). Avoid fixed survival thresholds or ablated percentages from this simple model.

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01. Bring back an earlier idea (retrieve) *Review: Unit 5, Day 7; 3 course days later.*
$Q=\Delta U+W_{by}=6$ J; heat enters.

02. Explain a worked example (worked)
Branches 700 and 300 J; could measure temperature change, emitted radiation or air heating with appropriate models.

03. Represent and interpret (model)
Smaller sphere has greater area per mass; stronger surface exposure per kg, all else equal.

04. Solve and justify (practice)
K 1.125×10^{9} J; requirement 8.6×10^{7} J; ratio 13.1. This does not determine survival.

05. Analyze the reasoning (reason)
Entry angle/speed profile, density/strength, fragmentation, shape, heat-transfer rate, atmosphere; accept relevant factors.

06. Choose a method (mixed) *Review: Unit 4, Day 20; 12 course days later.*
New speed is 1/4; $v_{escape}=\sqrt{(2\ GM/r)}$.

07. Exit check (exit)
$A/m\propto1/r$, so half. KE per mass depends on speed; survival also depends on transfer rates, fragmentation and conditions.

08. STRETCH: Apply it another way (stretch)
$\Delta T=20$ K; simplified lumped model omits gradients, ablation, radiation and material failure.

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